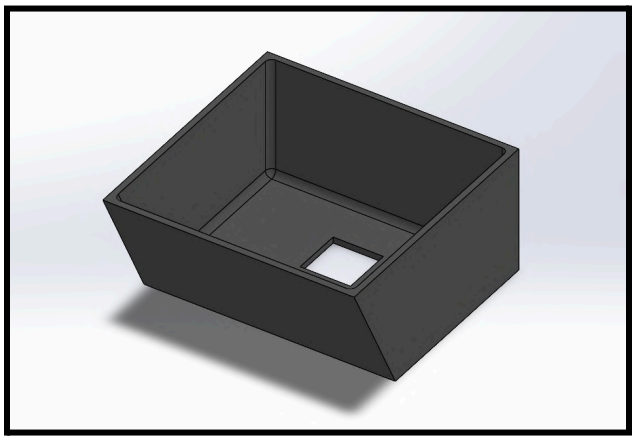
10. Technology selection for „make“ parts

10.1. Egg Storage

10.1.1. Egg Storage Container



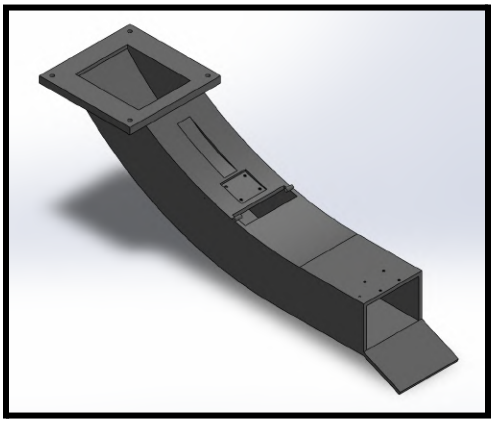
Material : PP-C HI / Polypropylene Copolymer

Injection moulding for the main raw egg storage container. The part is moulded as one PP-C plastic body, including the angled side walls, rounded internal fillets, reinforced top rim, and general outer shape.Moulded draft angles on the side walls.

A Core insert for the square middle cut-out. The square opening is formed directly in the injection mould using a fixed core insert, so it does not need to be cut out afterwards.
Stepped core pins The counterbore shape can be moulded directly using stepped core pins. If higher screw-seat accuracy is required, the holes are moulded slightly undersized and then finished using a drilling/counterboring operation.
Edge finishing using a manual deburring tool around the square cut-out, screw holes, top rim, and parting line. This removes plastic flash and sharp edges, making the part safer and easier to clean.

10.2. Egg Transportation

10.2.1. Egg Tunnel



Material name: PP-C HI / Polypropylene Copolymer

Manufacturing Technology:

Injection moulding as two PP-C half-shells for the curved egg transport tunnel. The tunnel is split into an upper half and lower half because a one-piece closed curved tube would be difficult to remove from the mould. The angled/curved egg path, rectangular tunnel shape, end flanges, and general outer profile are formed directly during moulding.

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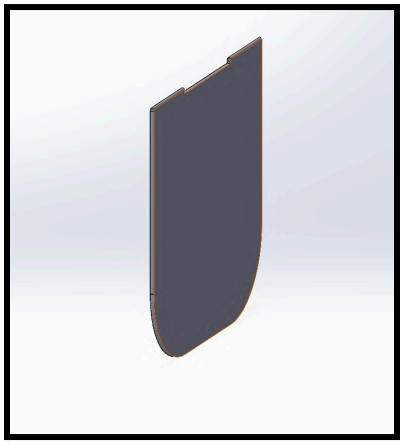
Steel injection mould with fixed core inserts to form the main internal egg path, square openings, end flanges, angled walls, and internal fillets. This gives a smooth food-contact surface and reduces secondary machining.
Draft angles on the tunnel walls and flange walls of approximately 1.5°–2° so both PP-C half-shells can be released from the injection mould without damage.

Side-action slides in the injection mould for the motor shaft hole and side sensor cut-outs. The slides move sideways before ejection, allowing the side holes and sensor windows to be moulded directly during production.

Core pins in the injection mould for the screw holes in the flanges. Hardened steel core pins form the screw clearance holes directly during moulding, so no drilling is required in normal production.

Ultrasonic welding machine used to join the upper and lower PP-C tunnel half-shells.
Heat-set threaded inserts The threaded fixing points are made by pressing brass heat-set inserts into moulded pilot holes after moulding. These inserts are placed in the flange and tunnel body where screws will be removed or tightened during assembly.
Mechanical deburring Done using a hand deburring tool around the drilled shaft hole, sensor cut-outs, square opening, and screw holes to remove burrs without creating sharp edges.

10.2.2. Gates for Transportation Tunnel



Material : POM-C / Acetal Copolymer

Justification: POM-C is selected because the gate is a rotating moving part and requires low friction, good stiffness, and good wear resistance at the shaft connection. It is preferred over PP because PP is lighter but softer and less wear-resistant. It is preferred over stainless steel because steel is unnecessarily heavy for this small low-load gate.

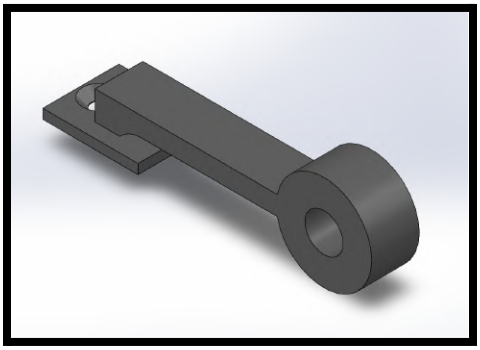
Manufacturing Technology:

Injection moulding for the rotating egg gate. The gate is moulded as one POM-C part, including the curved lower profile, flat gate face, top locating features, and general outer shape. The mould will be Polished Steel for a smooth surface.Draft angles of approximately 1°–2° on the gate faces and side edges so the gate can be released from the mould without damage.**Core pin in the injection mould for the motor shaft hole.** A hardened steel round core pin forms the shaft hole directly during moulding.

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Mechanical deburring around the parting line, shaft hole, top notches, and outer edges using plastic-safe deburring tools. This removes mould flash and sharp edges without damaging the food-contact surface.

10.2.3 Gate Shaft Holder for Tunnel



Material: POM-C / Acetal copolymer

Justification:
POM-C is used because the gate rotates inside the tunnel and needs low friction, good stiffness, and wear resistance around the shaft hole. It is preferred over PP because the hinge area and thin gate arm need better dimensional stability and lower wear.

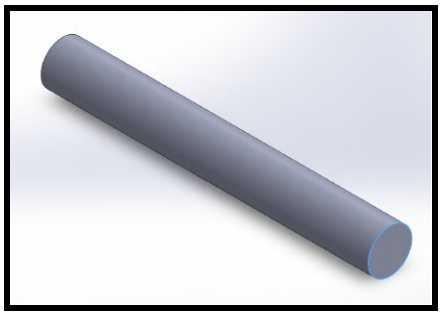
Manufacturing Technology:

- **Injection moulding:** The gate is moulded as one plastic part, including the cylindrical hinge hub, long gate arm, rectangular stop/actuator section, and flat rear plate.
- **Mould parting line:** The parting line is placed around the side edge of the flat gate arm and rear plate so the main sliding/contact faces stay clean.
- **Core pin for hinge hole:** The shaft hole through the cylindrical hub is formed by a core pin running through the hub axis. The hole is then reamed to final size so the gate rotates smoothly on the shaft.
- **Core pin for rear mounting hole:** The rear hole is formed through the flat end plate in the screw/fastener direction so it lines up with the actuator or linkage connection.
- **Draft direction:** Small draft is applied to the flat gate arm, rear plate, and hub side walls in the mould opening direction so the part releases without damaging the thin arm.
- **Gate position:** The injection gate is placed on the rear non-contact side of the part, away from the hinge hole and the tunnel-contact face.
- **Finishing:** The hinge hole, rear hole, parting line, gate mark, and outer edges are deburred after moulding.

Important manufacturing features:
The cylindrical hub is the critical feature because it controls the gate rotation. The hinge hole is finished after moulding to correct shrinkage and keep the shaft fit accurate. The parting line and gate mark are kept away from the sliding/contact face so the gate does not scrape inside the tunnel.

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10.2.4 Egg Tunnel Shaft



Material: EN 1.4301 / AISI 304 stainless steel precision round bar, Ø4 mm

Justification:
EN 1.4301 is used because the shaft is close to the egg tunnel and must resist humidity, raw egg residue, and cleaning. A Ø4 mm stainless shaft is stiff enough for the low gate load and gives a smooth rotating surface for the plastic gate hub.

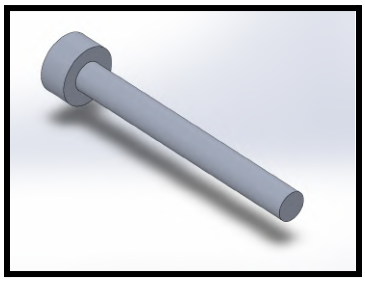
Manufacturing Technology:

- **Precision round bar cutting:** A Ø4 mm stainless steel precision ground rod is cut slightly longer than the final shaft length.
- **CNC turning / end finishing:** The shaft is clamped in a small collet chuck. Both ends are faced to the final 100 mm length and lightly chamfered so the shaft can slide through the gate hub and tunnel supports without damaging them.
- **Surface finishing:** The shaft diameter is polished if needed so the plastic gate rotates smoothly with low friction.
- **Deburring and passivation:** Both shaft ends and chamfers are deburred, then the shaft is passivated to restore corrosion resistance.

Important manufacturing features:
The Ø4 mm outer diameter is the functional bearing surface, so it must stay smooth and free from clamp marks. The end chamfers are important because the shaft passes through the plastic gate hub and should not scrape or crack the plastic during assembly. No threads are needed on the shaft itself; axial fixing can be done with small collars or clips outside the gate.

10.3. Egg Breaking Mechanism

10.3.1. Egg holding Shaft



Material: EN AW-6082 T6
Justification:
EN AW-6082 T6 is selected because the shaft rotates with the servo and must stay lightweight over its 200 mm length. Aluminium reduces rotating mass compared with stainless

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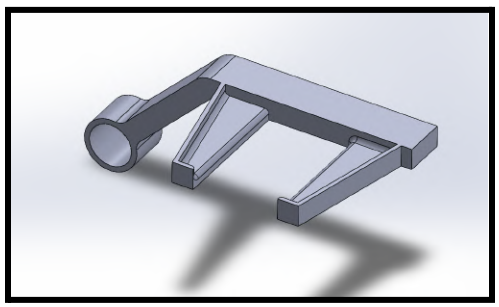
steel, while still being stiff enough for the egg holder load. It is also more suitable than plastic because the servo connection and long shaft need better stiffness and wear resistance.

Manufacturing Technology:

- **Automatic bar saw cutting:** The EN AW-6082 T6 round bar is cut slightly longer than the final shaft length to leave machining allowance on both ends.
- **Setup 1 - CNC turn-mill centre:** The raw bar is clamped in a collet chuck on the circular-head side, with extra stock left for gripping. The free shaft end is supported by a tailstock centre. In this setup, the long shaft diameter, shoulder, rear shaft end, and shaft-end chamfer are turned.
- **Setup 2- re-clamping in soft collet:** The part is flipped and clamped on the finished shaft diameter using a soft collet or protective split sleeve to avoid marking the shaft surface. The circular head face is then faced to final thickness.
- **C-axis milling on the head face:** The turn-mill C-axis locks the circular head in position. The servo horn pocket is milled into the front face of the circular head so the horn sits flat and aligned.
- **Hole machining on the head face:** The servo screw positions are spot drilled, drilled, and thread formed from the same head-face setup, so the pocket and screw holes stay aligned.
- **Final finishing:** The shaft end, shoulder, circular head edge, servo pocket, and threaded holes are deburred. The part is then clear anodised for corrosion resistance and cleanability.

Important manufacturing features:
The long shaft is supported during turning to prevent bending and vibration. The second setup clamps on the shaft diameter using soft tooling, not directly on the servo horn pocket or finished head face. The servo pocket and screw holes are machined in one setup on the circular head face to keep the servo horn accurately aligned with the shaft axis.

10.3.2. Egg Holding Arm



Material: EN AW-6082 T6

Justification:
EN AW-6082 T6 is used because the arm rotates with the shaft, so it must be lightweight but still stiff enough at the fork arms and ribs. Aluminium is preferred over stainless steel because it

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reduces servo load, and it is preferred over plastic because the hub and long arms need better stiffness and dimensional stability.

Manufacturing Technology:

Automatic band saw cutting to cut an EN AW-6082 T6 aluminium billet slightly larger than the final egg holder size.

5-axis CNC milling to machine the main arm shape, rectangular crossbar, open middle area, fork-like arms, triangular ribs, and outer profile. A 3-flute carbide end mill for aluminium is used for rough milling and profile cutting.

Finish milling on the 5-axis CNC milling centre to create the final egg-contact surfaces, fork-like tips, rib faces, and rounded edges. A 2-flute carbide ball-nose end mill for aluminium is used for the rounded surfaces and internal radii.

Circular milling of the cylindrical hub on the 5-axis CNC milling centre. A 3-flute carbide end mill for aluminium is used with circular interpolation to machine the outside diameter of the hub.

Spot drilling of the shaft hole position through the cylindrical hub using a carbide spot drill. This keeps the shaft hole centred and prevents the drill from walking.

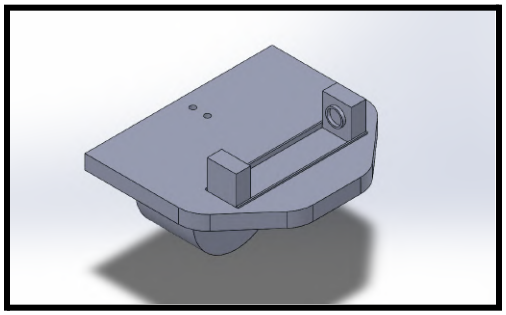
Drilling of the shaft hole through the hub using an HSS twist drill for aluminium.

Reaming of the shaft hole using a machine reamer to achieve an accurate fit between the holder and the rotating shaft.

c**lear anodising after machining** to improve corrosion resistance and cleanability in the raw egg/kitchen environment.

Final mechanical deburring using a hand deburring tool and fine abrasive pad around the shaft hole, fork tips, ribs, open middle area, and outer edges.

10.3.3. Base of Spear



Material:Clear anodised EN AW-6082 T6

Justification:

EN AW-6082 T6 is used because the base must keep the spear guide and shaft support accurate while staying lightweight. Clear anodising is added for corrosion resistance and easier cleaning in the raw egg area.

Manufacturing Technology:

Band saw cutting: An oversized aluminium billet is cut with extra stock for machining and clamping.

Setup 1 — top-side CNC milling: The billet is clamped from the bottom. The top reference face, guide rails, spear guide slot, rectangular cut-outs, curved outer profile, and small vertical mounting holes are machined from the top side.

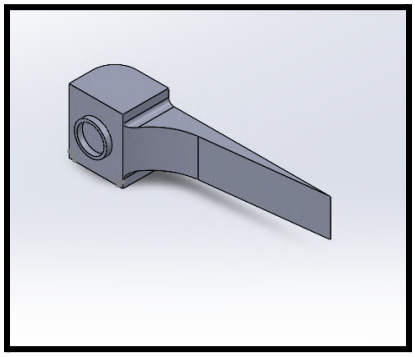
5-axis side machining: The machine indexes to the upright support block. The shaft/bushing hole is drilled and bored through the upright block in the shaft direction, so the spear shaft aligns with the guide slot.

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Setup 2 — bottom-side finishing: The part is flipped and held on the finished top face using soft jaws. The bottom reference face and lower rounded body are finished so the base sits flat in the assembly.
Finishing: The guide slot, mounting holes, shaft/bushing hole, rectangular cut-outs, upright block edges, and outer profile are deburred. The part is then clear anodised.

Important manufacturing features: The top face is the main datum because it controls the spear guide slot and mounting position. The shaft/bushing hole is machined after the upright block is finished so it stays aligned with the spear movement. The bottom face is finished last to make the base sit flat during assembly.

10.3.4. Spears : Left and Right



Material: Martensitic stainless steel, EN 1.4116 / X50CrMoV15, DIN EN 10088-3
Justification:
EN 1.4116 is used because the spear needs a hard, sharp cracking edge that can repeatedly contact eggshells without bending or wearing quickly. It is preferred over aluminium and AISI 304 because they are too soft for the cracking edge, while EN 1.4116 can be hardened, ground, polished, and passivated for the raw egg area.

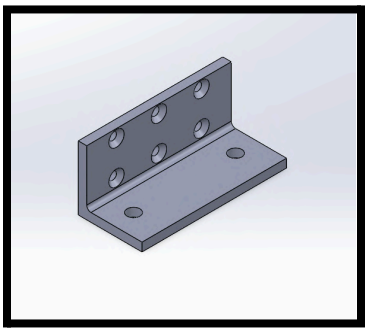
Manufacturing Technology:
Band saw cutting: Oversized EN 1.4116 billets are cut for the left and right mirrored spears, leaving machining allowance on the mounting block and tapered blade.
Setup 1 — mounting block milling: The billet is clamped on the bottom face. The rear mounting block, flat reference faces, curved transition, and rough tapered spear profile are CNC milled from the top side.
Hole and counterbore machining: The mounting block remains in the same setup. The shaft hole is spot drilled, drilled, bored, and reamed through the block. The counterbore is machined on the outside face of the block for the bushing or shaft seat.
Setup 2 — taper and underside finishing: The part is flipped and held on the finished mounting block faces using soft jaws. The underside of the spear body and final tapered blade thickness are milled.
Cracking edge grinding: The spear tip and lower cracking edge are ground along the long tapered blade. This edge is kept sharp and is not rounded during deburring.

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Heat treatment and finish grinding: The spear is vacuum hardened and tempered, then the cracking edge and shaft hole area are finish ground to restore final accuracy after heat treatment.
Finishing: Only the non-cutting edges, mounting block edges, counterbore edge, and shaft hole edge are deburred. The spear is electropolished and passivated for corrosion resistance and cleanability.

Important manufacturing features:
The mounting block is the main datum because it controls the shaft hole and spear position in the mechanism. The shaft hole and counterbore are machined from the same setup to keep them concentric. The cracking edge is ground after milling and protected during final deburring so the spear remains sharp.

10.3.5 Solenoid Motor Mounting Bracket



Material: EN 1.4301 / AISI 304 stainless steel sheet, 3.0 mm

Justification:
EN 1.4301 is used because the bracket is inside the machine and must resist humidity, oil vapour, and cleaning. A 3.0 mm stainless sheet gives enough stiffness to hold the solenoid motor and its casing without making the bracket unnecessarily heavy.

Manufacturing Technology:

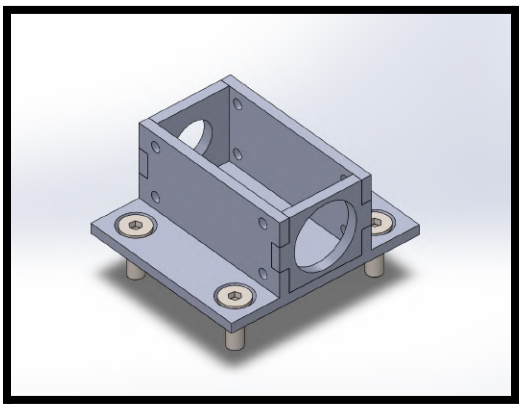
- **CNC fibre laser cutting:** The flat blank is cut from stainless steel sheet, including the outer profile, base mounting holes, vertical solenoid/casing holes, and bend reliefs.
- **Hole cutting:** The base holes and vertical flange holes are cut while the sheet is flat so the hole pattern is positioned accurately before bending.
- **Press brake bending:** The sheet is bent 90° along the bend line between the horizontal base flange and the vertical mounting flange. The inside radius is created by the press brake tool.
- **Bend direction:** The vertical flange is bent upward from the base flange. The base holes fix the bracket to the machine frame, while the vertical hole pattern mounts the solenoid motor and casing.
- **Hole finishing:** The hole edges are deburred after bending, especially around the vertical solenoid mounting holes and base holes, so the screws sit flat and the casing is not scratched.

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- **Surface finishing:** The outer edges, bend reliefs, hole edges, and flange corners are rounded. The part is brushed and passivated after cutting and bending.

Important manufacturing features:
The bend line controls the 90° angle between the base and vertical mounting face, so it controls the solenoid alignment. The vertical hole pattern is the critical feature because it locates the solenoid motor and casing. The base holes stay on the horizontal flange so the bracket can be fixed securely to the frame.

10.3.6 Solenoid Motor Casing



Material: Clear anodised EN AW-6082 T6 aluminium

Justification:
EN AW-6082 T6 is used because the casing must hold the solenoid motor accurately while staying lightweight and stiff. Aluminium is suitable because the part is not in direct food contact, is easy to machine, and provides enough strength for the solenoid mounting and frame connection.

Manufacturing Technology:

- **Block cutting:** An oversized aluminium billet is cut to leave machining allowance for the base flange, side walls, and front/rear faces.
- **Setup 1 — base and top machining:** The billet is clamped from the outer side faces. The bottom mounting face, base flange, open internal cavity, wall thickness, and top rim are CNC milled from the top side.
- **Large circular openings:** The front and rear circular solenoid openings are drilled and bored through the vertical end faces. These openings are machined from the solenoid axis direction so the motor body sits straight inside the casing.
- **Side holes and cut-outs:** The smaller side holes are drilled from the outside faces for fastening the solenoid cover, cable routing, or side-mounted components.
- **Base mounting holes:** The flange holes are drilled and counterbored from the top side of the base flange so the screw heads sit flat when the casing is fixed to the bracket or frame.

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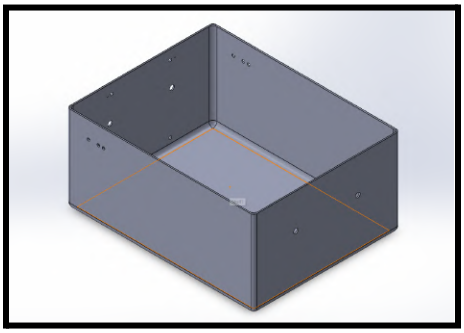
- **Setup 2 — bottom finishing:** The part is flipped and clamped on the finished top rim or side faces using soft jaws. The underside of the base is face milled so the casing sits flat on the mounting surface.
- **Finishing:** The large circular openings, side holes, base counterbores, top rim, internal cavity edges, and outside edges are deburred. The part is clear anodised after machining.

Important manufacturing features:
The bottom face is the main datum because it controls how the casing sits on the bracket. The large circular openings are the critical features because they locate the solenoid motor axis. The base counterbores are placed on the top side of the flange so the fixing screws sit flush and do not interfere with the solenoid body.

10.4. Egg Shell Storage

10.4.1. Egg Shell Storage (3 step manufacturing)

1.Main Drawer Box



Material: EN 1.4301 / X5CrNi18-10 stainless steel sheet, 1.0 mm
Justification:
EN 1.4301 is used because the drawer is exposed to wet eggshells, raw egg residue, cleaning water, and kitchen humidity. It gives enough corrosion resistance and cleanability for a non-direct food-contact storage part.

Manufacturing Technology:

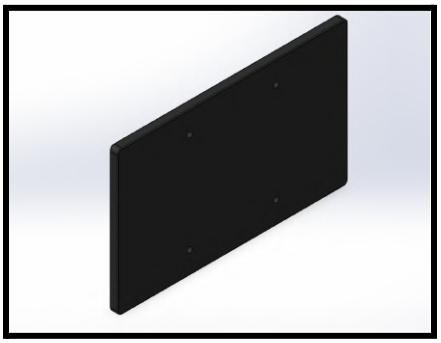
CNC fibre laser cutting: The flat sheet blank is cut with the outer profile, corner reliefs, shaft holes, handle holes, drainage/side holes, and top-edge geometry before bending.
Press brake bending: The bottom sheet is bent upward along the four drawer bend lines to form the front, rear, and side walls. The bend sequence starts with the shorter side walls, then the longer front and rear walls to avoid collision with the press brake tooling.
Top rim hemming: The upper edges of the drawer walls are folded outward/downward to remove sharp sheet edges and stiffen the open top rim.
TIG welding: The vertical corner seams are welded after bending to close the drawer corners and prevent egg residue from collecting in open gaps.
Weld finishing: The welded corner lines are ground smooth, especially on the inside

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corners, so the drawer can be cleaned easily.
Deburring and brushing: The laser-cut holes, top rim, front opening, welded corners, and outer edges are deburred. The visible stainless surfaces are brushed for a clean finish.

Important manufacturing features:
The bend lines define the drawer walls and must match the corner reliefs so the corners close cleanly after bending. The TIG weld seams are placed only on the vertical drawer corners. The inside welds are smoothed because this is where wet eggshell residue can collect.

2.Drawer Front Cover



Material: ABS-HI plastic
Justification:
ABS-HI is used because the front cover is a visible, non-food-contact part that needs good impact resistance, a clean surface finish, and enough stiffness around the screw holes. It is preferred over PP because it gives a better visible surface and is more dimensionally stable for a flat cover plate.

Manufacturing Technology:

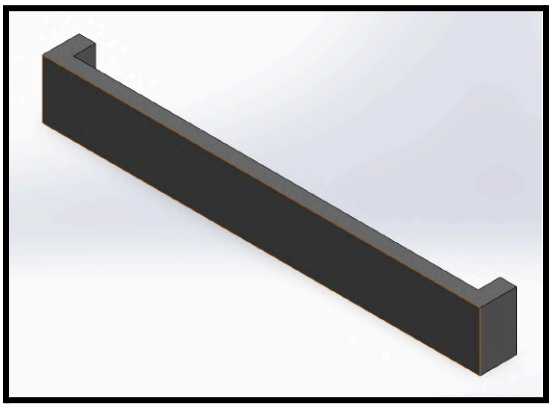
Injection moulding: The cover is moulded as one ABS part, including the flat front panel, rounded outer corners, raised/thickened border, and screw-hole bosses on the rear side.
Mould parting line: The parting line is placed around the outer edge of the cover so it is not visible on the main front face.
Core pins for holes: The visible screw holes and smaller locating holes are formed with core pins from the front-to-back direction. This keeps the holes aligned with the drawer mounting points.
Rear-side bosses: The area around the screw holes is locally thickened on the rear side to prevent cracking when the cover is screwed onto the drawer.
Draft direction: A small draft angle is applied on the outer edge and rear bosses in the mould opening direction so the cover can be released cleanly.

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Gate position: The injection gate is placed on the rear side of the cover so no gate mark appears on the visible front face.
Finishing: The outer edge, hole edges, and parting-line flash are trimmed and deburred after moulding.

Important manufacturing features:
The front face is kept free of gates, ejector marks, and parting lines because it is the visible user-facing surface. The holes are moulded in the front-to-back direction to match the screw direction during assembly. The rear bosses support the screw holes so the plastic cover does not deform or crack during tightening.

3.Drawer Handle



Material: ABS-HI plastic with brass threaded inserts

Justification:
ABS-HI is used because the handle is a visible user-contact part that needs good impact resistance, a clean surface finish, and low weight. Brass inserts are used for the threaded holes because repeated screw tightening would wear plastic threads.

Manufacturing Technology:

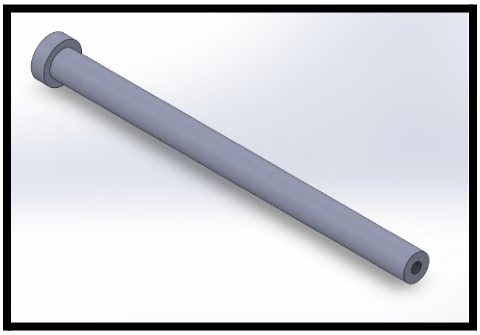
Injection moulding: The handle is moulded as one plastic part, including the long grip bar, thicker end blocks, rounded outer edges, and rear mounting bosses.
Mould parting line: The parting line is placed around the side edge of the handle so the front grip face stays clean and visible.
Core pins for mounting holes: The mounting holes are formed from the rear side of the handle, in the same direction as the screws used to attach it to the drawer cover.
Heat-set threaded inserts: Brass threaded inserts are pressed into the rear mounting holes after moulding. These create stronger threads for the drawer screws.
Draft direction: Small draft is applied on the side walls of the long grip and end blocks so the handle can release from the mould without marking the front face.
Gate position: The injection gate is placed on the rear side of the handle, away from

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the visible front grip surface.
Finishing: The parting-line flash, hole edges, and insert openings are trimmed and deburred after moulding.

Important manufacturing features:
The front grip face is kept free from gate marks and ejector marks because it is visible to the user. The threaded inserts are placed only in the rear mounting holes, where the handle is screwed to the drawer. The thicker end blocks support the inserts and prevent cracking when the handle is pulled.

4.Inner Shaft



Material: EN 1.4301 / X5CrNi18-10 stainless steel round bar

Justification:
EN 1.4301 is used because the shaft is inside the eggshell drawer and is exposed to wet eggshells, raw egg residue, and cleaning water. Stainless steel keeps the sliding surface corrosion-resistant and easy to clean.

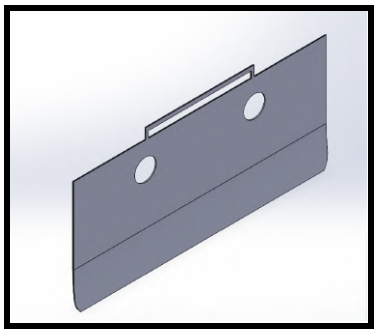
Manufacturing Technology:

- **Bar cutting:** The stainless steel round bar is cut slightly longer than the final shaft length to leave machining allowance on both ends.
- **Setup 1 — CNC turning:** The bar is clamped in a collet chuck and supported by a tailstock centre because the shaft is long. The long shaft diameter, end flange/stop collar, shoulder, and outer chamfers are turned in this setup.
- **Setup 2 — end machining:** The shaft is flipped and clamped on the finished shaft diameter using a soft collet. The end face is finished, then the axial hole is drilled and tapped from the shaft end.
- **Finishing:** The shaft ends, shoulder, flange edge, and threaded hole edge are deburred. The shaft surface is polished so the compression plate can slide smoothly.

Important manufacturing features:
The long shaft diameter is the functional sliding surface, so it is protected during re-clamping. The end thread is machined along the shaft axis so the fixing screw pulls straight into the shaft. The flange/stop collar controls the shaft position inside the drawer.

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5.Manual Compression Plate



Material: EN 1.4301 / X5CrNi18-10 stainless steel sheet

Justification:
EN 1.4301 is used because the plate moves inside the eggshell drawer and is exposed to wet eggshells, raw egg residue, and cleaning water. It gives enough stiffness, corrosion resistance, and cleanability for the sliding compression part.

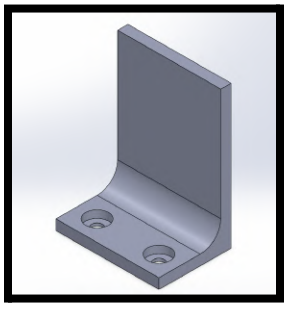
Manufacturing Technology:

- **CNC fibre laser cutting:** The flat sheet profile is cut first, including the outer contour, two shaft holes, long upper slot, top notch, and rounded lower corners.
- **Press brake bending:** The plate is bent along the horizontal bend line across the lower section. The bend gives the lower face its angled compression surface while keeping the shaft holes and upper slot in the flat upper area.
- **Hole finishing:** The two shaft holes are reamed after cutting so the plate can slide on the inner shafts without too much play.
- **Deburring and brushing:** The shaft holes, slot, notch, outer profile, rounded corners, and bend edge are deburred. The plate faces are brushed to improve cleanability.
- **Passivation:** The plate is passivated after cutting, bending, and finishing to restore corrosion resistance.

Important manufacturing features:
The shaft holes are kept in the flat upper section because they guide the plate along the drawer shafts. The bend line is placed below the shaft holes so the lower plate area can compress the eggshells without disturbing the sliding fit. All cut edges are deburred because the part is handled and cleaned regularly.

10.5. Oil Storage

10.5.1 Oil Storage Mount



Material: EN AW-6082 T6 aluminium
Justification:
EN AW-6082 T6 is used because the bracket must hold the oil system rigidly while staying lightweight. Aluminium is suitable here because the part is not in direct food contact, is easy to machine, and provides enough stiffness for the mounting load.

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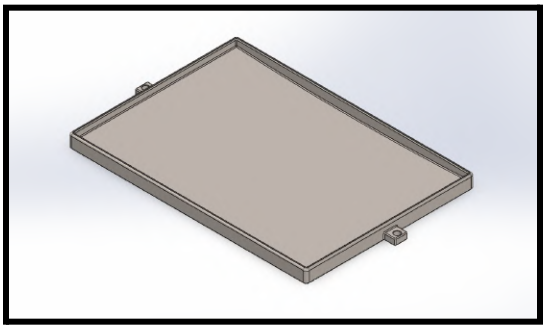
Manufacturing Technology:

- **Band saw cutting:** An oversized aluminium billet is cut to leave machining allowance for the L-shape, vertical wall, and mounting base.
- **Setup 1 — bottom/base machining:** The billet is clamped from the side faces. The bottom mounting face, outer base profile, and two counterbored mounting holes are machined from the top side. The counterbores are placed on the upper face so the screw heads sit flush.
- **Setup 2 — upright wall machining:** The part is rotated and clamped on the finished base face using soft jaws. The vertical mounting face, top edge, and side faces of the upright wall are milled.
- **Inside radius milling:** The curved internal transition between the base and upright wall is milled with a ball-nose end mill. This radius reduces stress concentration where the oil system load bends the bracket.
- **Finishing:** The counterbore edges, screw holes, inner radius, upright wall edges, and outer profile are deburred. The part can be clear anodised for corrosion resistance and cleanability.

Important manufacturing features:
The bottom face is the main datum because it controls how the bracket sits on the machine frame. The vertical wall face is machined perpendicular to the base so the oil system sits straight. The screw holes and counterbores are machined from the mounting side to keep the screw heads flush and accessible during assembly.

10.6. Conveyor System

10.6.1. Pocket Plate



Material: EN 1.4409 / GX2CrNiMo19-11-2 stainless steel precision casting

Justification:
EN 1.4409 is used because the plate is in direct contact with raw egg, oil, heat, and cleaning water. It gives high corrosion resistance and hygiene performance. Casting is preferred because the concave pocket, raised walls, rounded corners, and side tabs can be made as one part without welded joints in the food-contact area.

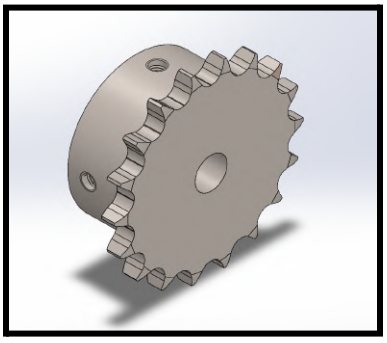
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Manufacturing Technology:

- **Wax pattern and investment casting:** The concave egg pocket, raised rim, rounded internal corners, side walls, underside shape, and integrated chain-mounting tabs are cast as one body.
- **Gate placement:** Casting gates are placed on the underside and outer non-food-contact areas, so the top egg-contact pocket remains uninterrupted.
- **Gate removal and blasting:** The gate marks are cut off from the underside. The cast surface is shot blasted to remove shell residue before machining.
- **CNC finish machining:** The plate is clamped on the underside and outer side walls. The concave pocket, upper rim, side walls, and underside reference face are finish milled so the egg-contact surface is smooth and the plate sits flat.
- **Tab hole drilling:** The mounting holes are drilled through the integrated side tabs for the plate-to-chain fasteners.
- **Polishing and passivation:** The concave pocket, raised rim, side walls, and reachable outer surfaces are polished. The part is passivated after machining and polishing.

Important manufacturing features:
The top concave pocket is the main food-contact surface, so casting gates and clamping marks are kept away from it. The side tabs are cast together with the plate to avoid welded joints near the cooking area. The underside is machined as the reference surface so the plate mounts flat on the conveyor attachments.

10.6.2. Sprocket for Conveyor Belt



Sprocket for Conveyor Belt
Material: EN 1.4301 / X5CrNi18-10 stainless steel round bar

Justification:
EN 1.4301 is used because the sprocket is close to the food and heating area, so it must resist oil, humidity, cleaning water, and corrosion. It also has enough strength and wear resistance to drive the 06B-1 roller chain.

Manufacturing Technology:

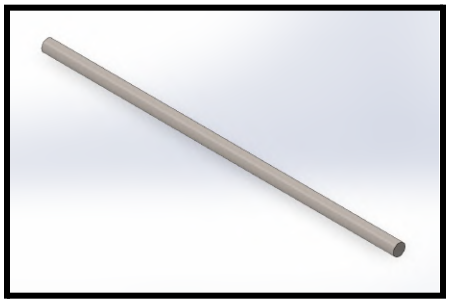
- **Band saw cutting:** A stainless steel round-bar blank is cut slightly thicker than the final sprocket width.
- **Setup 1 — CNC turning:** The blank is clamped in a chuck. Both side faces, the hub step, outside blank diameter, and central shaft bore are turned from the same axis so the bore and sprocket body stay concentric.

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- **Bore finishing:** The central bore is bored and reamed to the final shaft fit. This bore becomes the reference for the later tooth and locking-hole operations.
- **Sprocket tooth cutting:** The part is mounted on a mandrel through the finished central bore. The 06B-1 tooth profile is hobbed around the outside diameter so the teeth are concentric with the shaft bore.
- **Locking screw holes:** The sprocket is held on the central bore and indexed on a 4-axis CNC machine. The radial locking holes are drilled and tapped through the hub side so the sprocket can be fixed to the shaft.
- **Finishing:** The tooth edges, bore edges, side faces, hub edges, and threaded-hole openings are deburred. The visible and cleanable surfaces are polished, then the sprocket is passivated.

Important manufacturing features:
The central bore is the main datum because it controls sprocket rotation on the shaft. The teeth are cut after the bore is finished so the chain pitch circle stays concentric with the shaft. The locking screw threads are placed radially through the hub, not through the teeth, so the screws clamp directly onto the shaft without weakening the tooth profile.

10.6.3. Driving Shaft/Idler Shaft (Same Technology just one is slightly longer)



Material: EN 1.4301 / X5CrNi18-10 stainless steel round bar

Justification:
EN 1.4301 is used because the shafts are close to the food, oil, heat, and cleaning area. It provides corrosion resistance and enough strength to support the sprockets and keep the conveyor chains aligned.

Manufacturing Technology:

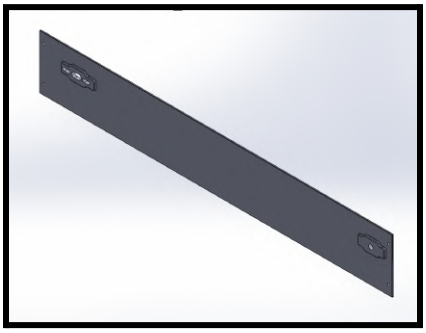
- **Bar cutting:** The stainless steel round bar is cut slightly longer than the final shaft length.
- **Setup 1 — CNC turning:** The bar is clamped in a collet chuck and supported with a tailstock centre because the shaft is long and slender. The shaft diameter, end face, and first end chamfer are turned from this setup.

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- **Setup 2 — end finishing:** The shaft is flipped and clamped on the finished diameter using a soft collet. The second end face is turned to final length and the second chamfer is added.
- **Bearing and sprocket seating surface:** The full shaft diameter is centreless ground or polished so the bearings and sprockets sit accurately and slide on without damaging the surface.
- **Finishing:** Both shaft ends, chamfers, and bearing/sprocket seating areas are deburred and passivated.

Important manufacturing features:
The shaft diameter is the main functional surface because it carries the bearings and sprockets. Soft clamping is used after turning so the finished shaft surface is not marked. The end chamfers help the shaft pass through the bearings and sprocket bores during assembly.

10.6.4. Conveyor Side Frame



Material: EN 1.4301 / X5CrNi18-10 stainless steel plate
Justification:
EN 1.4301 is used because the side frame is close to the conveyor, oil, heat, and cleaning area. It provides enough stiffness for the shafts and bearing mounts while giving suitable corrosion resistance for a non-direct food-contact structural part.

Manufacturing Technology:

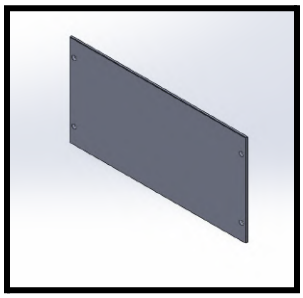
- **CNC fibre laser cutting:** The long rectangular outer profile, end mounting holes, bearing/shaft openings, and adjustment slots are cut from the stainless steel plate in the flat state.
- **Slot finishing:** The bearing adjustment slots are finished by CNC milling if tighter fit is needed. These slots allow the shaft/bearing position to be adjusted for chain tensioning.
- **Hole finishing:** The bearing mounting holes and frame fixing holes are drilled or reamed after laser cutting where accurate alignment is required.
- **Edge chamfering:** The long outer edges, slot edges, shaft openings, and mounting holes are lightly chamfered to remove sharp laser-cut edges.
- **Surface finishing:** The visible faces are brushed, and the part is passivated after cutting and machining.

Important manufacturing features:
The two end slot areas are the critical features because they control shaft and bearing

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alignment. The holes and slots are made while the part is flat so their spacing stays accurate along the full frame length. No bending is used for this part, so the flatness of the plate must be maintained during cutting and finishing.

10.6.5. Conveyor Back Frame



Material: EN 1.4301 / X5CrNi18-10 stainless steel sheet

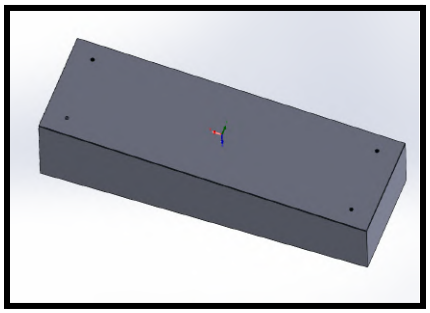
Justification:
EN 1.4301 is used because the back frame supports the conveyor structure and keeps the two side frames aligned. It is close to heat, oil, humidity, and cleaning water, so stainless steel is suitable for corrosion resistance and cleanability.

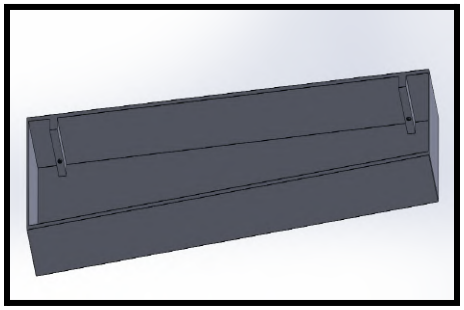
Manufacturing Technology:

- **CNC fibre laser cutting:** The rectangular outer profile and corner mounting holes are cut from a flat stainless steel sheet in one setup.
- **Mounting hole finishing:** The corner holes are drilled or reamed after laser cutting if tighter clearance is needed for the frame fasteners.
- **Edge finishing:** The outer profile and mounting holes are deburred and lightly chamfered to remove sharp laser-cut edges.
- **Surface finishing:** The visible faces are brushed, and the part is passivated after cutting and finishing.

Important manufacturing features:
The mounting holes are the critical features because they locate the back frame between the two conveyor side frames. All holes are made while the sheet is flat so the spacing stays accurate. No bending or welding is used, so the part must remain flat to keep the side frames aligned.

10.6.6. Frame under conveyor





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Conveyor Belt Hollow Support Frame

Material: EN AW-6082 T6 aluminium billet

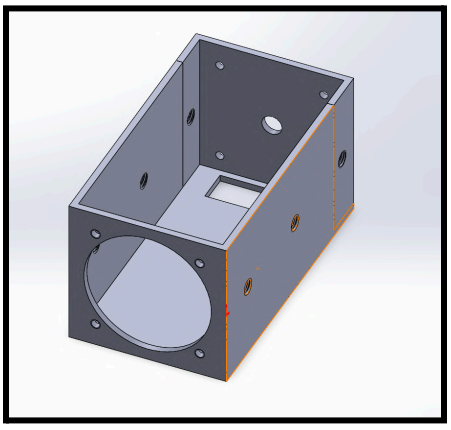
Justification:
EN AW-6082 T6 is used because the frame must support the conveyor while staying lightweight and stiff. The solid billet allows the underside to be hollowed out while leaving thick bosses for the M10 threaded mounting points.

Manufacturing Technology:

- **Block sawing:** An oversized aluminium billet is cut to leave machining allowance on the outer walls, top face, and bottom pocket.
- **Setup 1 — top-side machining:** The billet is clamped from the side faces. The top reference face, outer rectangular profile, and three M10 hole positions are machined from the top side.
- **M10 hole machining:** The three holes are drilled vertically through the top face using Ø8.5 mm core holes, then tapped M10×1.5 from the same top-side setup. This keeps the threads perpendicular to the conveyor mounting surface.
- **Setup 2 — underside pocket machining:** The part is flipped and clamped on the finished outer side faces using soft jaws. The underside cavity is milled out, while thicker internal bosses are left directly under the M10 threaded holes.
- **Edge finishing:** The outer profile, top edge, underside pocket edge, internal bosses, and M10 hole openings are chamfered and deburred.
- **Cleaning:** The hollow underside channel and threaded holes are washed to remove aluminium chips and cutting oil before assembly.

Important manufacturing features:
The top face is the main datum because it supports the conveyor assembly. The M10 threaded holes are machined before the underside pocket so their positions stay accurate. The underside pocket removes weight, while the internal bosses remain under the threaded holes to prevent thread stripping and local deformation during tightening.

10.6.7. Nema 17/Gearbox/Encoder Motor Housing



Material: EN AC-43500 / EN AC-AlSi10MnMg cast aluminium alloy

Justification:
EN AC-43500 is used because the housing must keep the NEMA 17 motor, gearbox, and encoder aligned

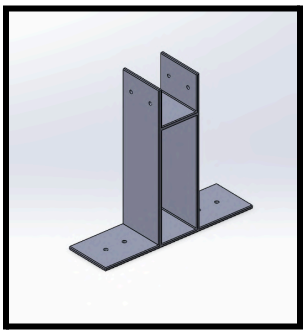
while remaining lightweight. Aluminium casting also allows the box shape, internal cavity, motor opening, and mounting features to be produced as one stiff casing.

Manufacturing Technology:

- **High-pressure die casting:** The housing body is cast with the outer box shape, open top, side walls, rear wall, front motor face, and internal cavity.
- **Parting line and draft:** The die-casting parting line is placed around the outer side of the housing. Draft is applied on the outer walls, inner cavity walls, and vertical ribs so the casing releases from the die.
- **Core inserts:** A main core insert forms the internal cavity for the motor, gearbox, and encoder. A second core forms the rough front gearbox opening.
- **CNC reference machining:** The front motor mounting face, top rim, bottom reference face, and side reference faces are milled after casting so the motor and gearbox sit square in the housing.
- **Front opening machining:** The front gearbox opening is bored to final diameter from the motor mounting face side, keeping it concentric with the motor position.
- **Mounting holes and threads:** The motor, gearbox, side-cover, rear-cover, and frame mounting holes are drilled and thread-formed after casting. These threaded holes are machined from their assembly side so screws enter straight during assembly.
- **Finishing:** The front opening, top rim, internal cavity edges, threaded holes, and outer edges are chamfered and deburred.
- **Powder coating:** The outside casing is powder coated for protection. The front motor face, bottom reference face, bored opening, and threaded holes are masked so coating does not affect assembly accuracy.

Important manufacturing features:
The front motor face is the main datum because it controls motor and gearbox alignment. The internal cavity is cast with a core insert instead of fully machined from solid material. Precision faces and threaded holes are machined after casting because cast surfaces alone are not accurate enough for the motor, gearbox, encoder, and cover alignment.

10.6.8 Nema Motor Mount



Material: EN 1.4301 / AISI 304 stainless steel sheet, 3.0 mm

Justification:
EN 1.4301 is used because the motor mount is close to the conveyor, oil vapour, humidity, and cleaning area. Stainless steel

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gives enough stiffness and corrosion resistance for holding the NEMA 17 motor and keeping the drive shaft aligned.

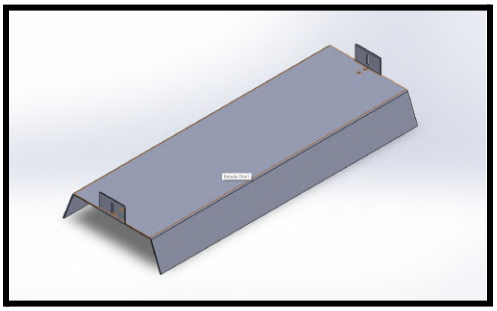
Manufacturing Technology:

- **CNC fibre laser cutting:** The base plate, two vertical motor support plates, and internal spacer/gusset plates are cut from stainless steel sheet. The base fixing holes, motor mounting holes, and motor shaft clearance opening are cut while the plates are flat.
- **Motor mounting hole finishing:** The NEMA 17 screw holes and central shaft opening on the vertical mounting plate are drilled or reamed after laser cutting if tighter alignment is required.
- **Welding fixture positioning:** The vertical plates are positioned perpendicular to the base plate in a welding fixture. The fixture keeps the motor face square to the base so the motor shaft aligns with the conveyor drive shaft.
- **TIG tack welding:** Tack welds are placed first at the base-to-vertical plate contact edges and at the spacer/gusset contact edges to lock the geometry before final welding.
- **TIG seam welding:** Weld seams are applied along the bottom edges of the vertical plates and along the spacer/gusset contact lines. These welds stiffen the motor mount against motor torque.
- **Edge finishing:** The base holes, motor holes, shaft opening, outer edges, and welded seams are deburred. The visible surfaces are brushed and passivated after welding.

Important manufacturing features:
The vertical motor plate is the critical reference because it controls the NEMA 17 motor position. The base plate holes fix the mount to the conveyor frame, while the motor holes locate the motor relative to the drive shaft. The spacer/gusset plates are welded between the vertical plates to reduce vibration and prevent the motor plate from bending under torque.

10.7. TOP IR System

10.7.1. TOP IR Shielding



Material: EN 1.4301 / X5CrNi18-10 stainless steel sheet, 2.0 mm

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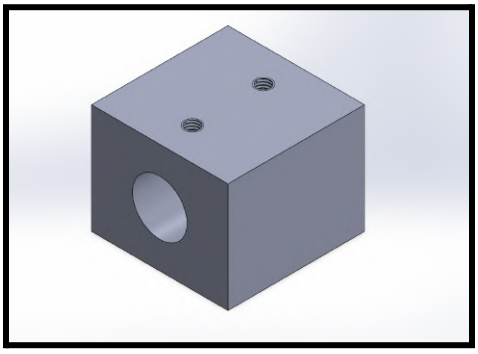
Justification:
EN 1.4301 is used because the shield is close to the IR heating zone and must resist heat, oil vapour, humidity, and cleaning. It is suitable because the part is a heat shield, not a direct food-contact cooking surface.

Manufacturing Technology:

- **CNC fibre laser cutting:** The flat sheet blank is cut with the outer profile, slotted mounting holes, circular screw hole, integrated tabs, bend reliefs, and front/back edge shape.
- **Slotted hole cutting:** The vertical slots are laser cut in the mounting tabs while the sheet is flat. These slots allow height adjustment when fixing the shield in the assembly.
- **Bend relief cutting:** Relief cuts are placed at the tab bases and side-wall transitions so the material does not tear or distort during bending.
- **Press brake bending — side walls:** The two long bend lines are bent downward to form the trapezoidal hood shape around the IR tube and guide the heat downward.
- **Press brake bending — mounting tabs:** The integrated tabs are bent upward from the top surface after the side walls are formed. The tabs are made from the same sheet, so no welded tab joints are needed.
- **Finishing:** The lower lips, slotted holes, circular hole, bend reliefs, tabs, and outer edges are deburred. The visible surfaces are brushed and then passivated.

Important manufacturing features:
The two long bend lines define the trapezoidal hood angle, so they control the final clearance around the IR tube. The tab bend lines control the mounting position of the shield. The bend reliefs are placed at the tab roots and side transitions to prevent cracking during press brake bending.

10.7.2. Ceramic Block for holding Quarts IR



Material: 96% aluminium oxide ceramic, Al_2O_3 C799

Justification:
Al_2O_3 C799 is used because the block sits directly beside the quartz IR tube and must resist high temperature while electrically insulating the tube from the metal shield. Stainless threaded inserts are used because ceramic threads would be too brittle for repeated screw tightening.

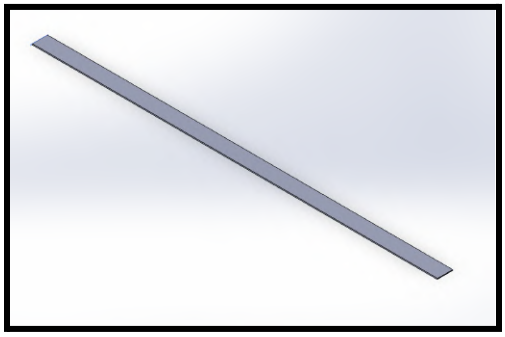
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Manufacturing Technology:

- **Ceramic powder pressing:** The rectangular block blank is pressed with extra allowance for sintering shrinkage.
- **Green-state machining:** The quartz-tube hole and top insert holes are machined while the ceramic is still in the softer green state. The tube hole is placed through the side of the block, while the insert holes are placed from the top face for the mounting screws.
- **Sintering:** The machined green block is fired in a ceramic furnace. The hole and outer dimensions include shrinkage allowance so the final ceramic block reaches the required size after firing.
- **Diamond finishing:** After sintering, the outer faces are diamond ground flat. The quartz-tube hole is diamond finished to its final diameter so the tube is supported smoothly without point contact.
- **Threaded insert placement:** A2 stainless threaded inserts are bonded into the top holes using high-temperature ceramic adhesive. These inserts provide the screw threads for mounting the ceramic block to the shield.
- **Edge finishing:** The block edges, tube-hole edges, and insert-hole edges are rounded with diamond abrasives to reduce chipping during assembly.
- **Cleaning:** The tube hole, insert holes, and outer surfaces are cleaned to remove ceramic grinding dust before assembly.

Important manufacturing features:
The side through-hole is the critical feature because it supports the quartz IR tube. The top insert holes are kept separate from the tube hole so the mounting screws do not load the quartz tube directly. All final sizing after sintering is done by diamond grinding because fired ceramic is too hard and brittle for normal drilling.

10.7.3. Side Lining for TOP IR



Material: EN 1.4301 / X5CrNi18-10 stainless steel sheet, 2.0 mm

Justification:
EN 1.4301 is used because the rail is close to the Top IR heating zone and must resist heat, oil vapour, humidity, and cleaning. It is suitable because the rail is a support part for the perforated mesh, not a direct food-contact surface.

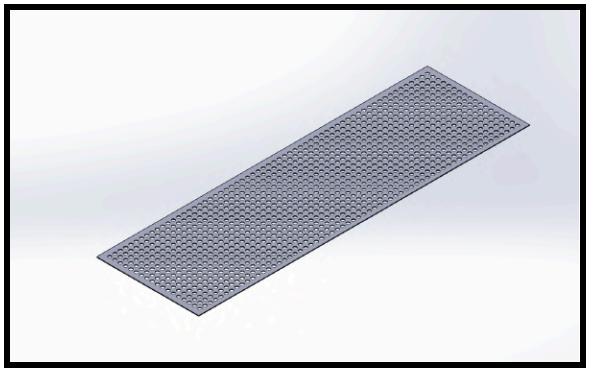
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Manufacturing Technology:

- **CNC fibre laser cutting:** The long flat strip, straight edges, square end, and slanted end are cut from stainless steel sheet in one flat operation.
- **Slanted end cutting:** The angled end is cut to match the sloped side wall of the Top IR shield, so the rail fits inside the trapezoidal hood without protruding.
- **No bending:** The part remains a flat strip. It has no bend lines because its function is only to act as an internal support ledge for the perforated mesh.
- **TIG tack welding:** The rail is first tack welded inside the Top IR shield at the required mesh height to hold its position before final welding.
- **TIG seam welding:** A weld seam is made along the contact edge between the rail and the inner wall of the Top IR shield. This fixes the rail permanently and creates a stable ledge for the mesh.
- **Finishing:** The long edges, slanted end, exposed square end, and welded edge are deburred. The weld area is cleaned and passivated after welding.

Important manufacturing features:
The slanted end follows the angled shield wall, so the rail sits flush inside the hood. The weld seam is placed along the rail-to-shield contact edge, not on the mesh-support surface. The rail must be aligned at the correct height before welding because it controls the final position of the perforated mesh.

10.7.4. Perforated Sheet



Material: EN 1.4301 / X5CrNi18-10 stainless steel perforated sheet, 1.0 mm
Justification:
EN 1.4301 is used because the mesh sits directly below the Top IR heater and must resist heat, oil vapour, humidity, and repeated cleaning. It is suitable because the part acts as a removable heat-protection and ventilation sheet, not as a direct cooking plate.

Manufacturing Technology:

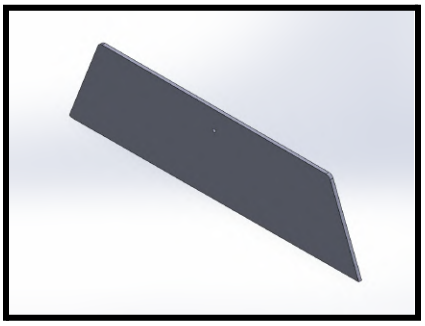
- **CNC turret punching:** The elongated ventilation slots are punched into the flat stainless steel sheet using an obround punch tool. The slot rows are kept evenly spaced so the sheet remains stiff and allows heat to pass through evenly.

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- **CNC fibre laser cutting:** After punching, the outer rectangular profile is laser cut to the final size. The outer border is aligned with the perforation field so the sheet fits correctly into the side lining rails.
- **Flatness correction:** The punched and cut sheet is roller-levelled to reduce warping caused by the large number of slots.
- **Deburring and edge rounding:** The slot edges and outer perimeter are deburred so the sheet can be handled and removed safely during cleaning.
- **Surface finishing:** The visible surface is brushed, and the sheet is passivated after punching, cutting, and deburring.

Important manufacturing features:
The slot pattern is the critical feature because it controls heat passage and prevents large food or shell pieces from reaching the IR tube. The outside perimeter must stay straight because the sheet slides into the welded side lining rails. No welding is used because the mesh must remain removable and replaceable for cleaning and maintenance.

10.7.5. TOP IR Back Frame



Material: EN 1.4301 / X5CrNi18-10 stainless steel sheet, 2.0 mm

Justification:
EN 1.4301 is used because the back frame is close to the IR heating zone and must resist heat, oil vapour, humidity, and cleaning. It also matches the Top IR shield material, which makes welding and corrosion protection consistent.

Manufacturing Technology:

- **CNC fibre laser cutting:** The flat back-frame profile is cut from stainless steel sheet, including the angled side edges, rounded corner transitions, lower edge, and central mounting hole.
- **Central hole cutting:** The central hole is laser cut in the same flat-sheet setup as the outer profile so it stays centred relative to the back-frame shape.
- **No bending:** The part remains a flat rear closing plate. It has no bend lines because it is welded directly to the rear opening of the Top IR shield.
- **TIG tack welding:** The back frame is tack welded to the rear of the Top IR shield first. It is aligned with the sloped side walls so the rear hood opening closes cleanly.

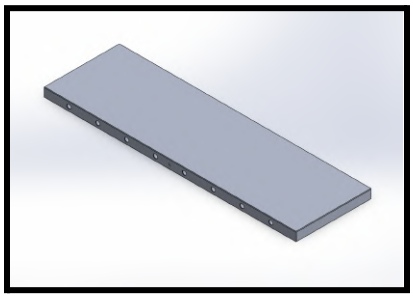
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- **TIG seam welding:** Weld seams are applied along the contact edges between the back frame and the Top IR shield. These seams permanently close the rear side and increase hood stiffness.
- **Finishing:** The angled edges, rounded corners, central hole, lower edge, and welded seams are deburred. The weld areas are cleaned, brushed, and passivated.

Important manufacturing features:
The angled side edges match the trapezoidal side walls of the Top IR shield, so the back frame fits without gaps before welding. The weld seams are placed around the rear contact edges, not across the central hole. The central hole is cut before welding because it is easier to position accurately while the sheet is flat.

10.8. Bottom Heating System

10.8.1. Heating block



Material: EN AW-6082 T651 aluminium plate

Justification:
EN AW-6082 T651 is used because the block must spread heat from the cartridge heaters evenly to the egg pocket plates. Aluminium is preferred over stainless steel because it has much better thermal conductivity and keeps the heating block lighter.

Manufacturing Technology:

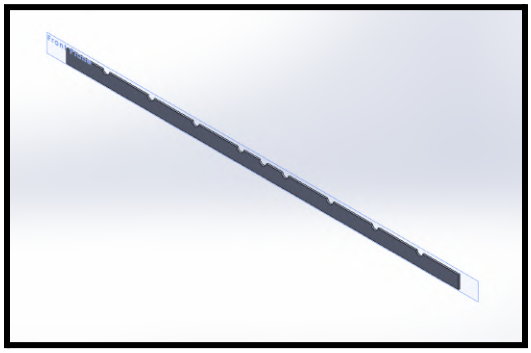
- **Band saw cutting:** An oversized aluminium plate blank is cut to leave machining allowance on the length, width, and thickness.
- **Setup 1 — top and outer machining:** The blank is clamped from the side faces. The top heating surface, rectangular outer profile, and side faces are CNC milled. The top surface is the main heat-transfer face, so it is finished flat.
- **Setup 2 — bottom reference machining:** The block is flipped and clamped on the finished side faces. The bottom reference surface is face milled so the block sits flat in the heating frame.

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- **Cartridge heater holes:** The Ø8 mm heater holes are deep-drilled from the long side face along the length of the block. The holes are kept parallel to the top heating surface so the heaters distribute heat evenly below the egg plates.
- **Temperature sensor hole:** The sensor hole is drilled from the side or bottom into the centre zone between the cartridge heater holes, so the sensor measures the actual block temperature instead of only the heater surface temperature.
- **Finishing:** The heater-hole openings, sensor hole, mounting edges, side edges, and machined corners are deburred. The top heating surface and visible side faces are brushed.

Important manufacturing features:
The top face is the critical heat-transfer surface and must remain flat and clean. The cartridge heater holes are drilled from the long side because the heaters are inserted lengthwise into the block. The sensor hole is positioned between heater holes so the temperature control represents the average heating block temperature.

10.8.2. Side Frame



Material: EN 1.4301 / X5CrNi18-10 stainless steel sheet, 2.0 mm

Justification:
EN 1.4301 is used because the strip is close to the bottom heating block and must resist heat, oil vapour, humidity, and cleaning. It is suitable because the strip is a support/clearance part, not a direct food-contact cooking surface.

Manufacturing Technology:

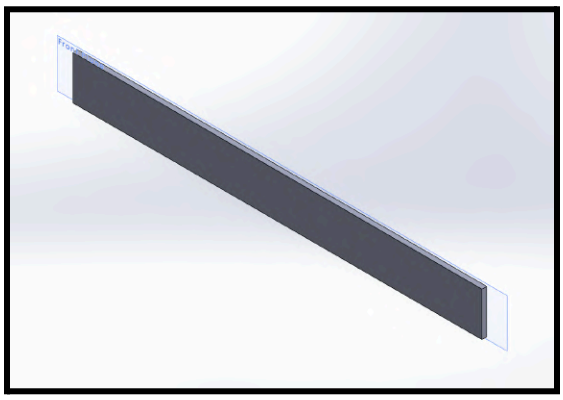
- **CNC fibre laser cutting:** The long strip profile, straight lower edge, end edges, and repeated semicircular notches are cut from flat stainless steel sheet.
- **Notch cutting:** The semicircular notches are laser cut along the upper edge with equal spacing. These notches clear or locate the repeated heater-related parts in the bottom heating assembly.
- **No bending:** The part remains a flat strip, so there are no bend lines or forming operations.
- **Edge finishing:** The long edges, end edges, and every semicircular notch are deburred and rounded so the strip does not damage nearby heater sleeves, wiring, or assembly parts.

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- **Surface finishing:** The visible faces are brushed, and the part is passivated after cutting and deburring.

Important manufacturing features:
The notch pattern is the critical feature because it must match the repeated heater positions. The strip is cut while flat so the notch spacing stays accurate. All notch edges are rounded because they are close to heater-related parts and wiring.

10.8.3. Front Frame



Material: EN 1.4301 / X5CrNi18-10 stainless steel sheet, 2.0 mm

Justification:
EN 1.4301 is used because the front frame is close to the bottom heating zone and must resist heat, oil vapour, humidity, and cleaning. It is suitable because the part is a welded support strip, not the main heat-transfer surface.

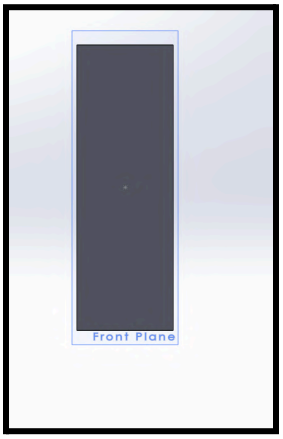
Manufacturing Technology:

- **CNC fibre laser cutting:** The flat front-frame strip is cut from stainless steel sheet, including the long edges and end edges.
- **No bending:** The part remains a flat strip, so there are no bend lines or forming operations.
- **Edge finishing:** The long edges and end edges are deburred and lightly rounded so the strip is safe to handle and fits cleanly against the side and bottom frame parts.
- **Welding fixture positioning:** The front frame is placed in a welding fixture together with the side frames and bottom frame. The fixture keeps the strips square and aligned before welding.
- **TIG tack welding:** Tack welds are placed at the corner/contact points first to hold the front frame in position.
- **TIG seam welding:** Weld seams are applied along the contact edges between the front frame, side frames, and bottom frame to create the welded bottom-heating frame.
- **Weld finishing:** The welded edges are cleaned, deburred, brushed, and passivated after welding.

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Important manufacturing features:
The front frame is cut as a flat strip and then becomes part of the welded heating-frame structure. The weld seams are placed only along the contact edges with the side and bottom frame parts. The welding fixture is important because it keeps the rectangular frame aligned before final welding.

10.8.4. Bottom Frame



Material: EN 1.4301 / X5CrNi18-10 stainless steel sheet, 2.0 mm

Justification:
EN 1.4301 is used because the bottom frame is close to the heating zone and must resist heat, oil vapour, humidity, and cleaning. It is suitable because the part is a welded support strip, not the main heat-transfer surface.

Manufacturing Technology:

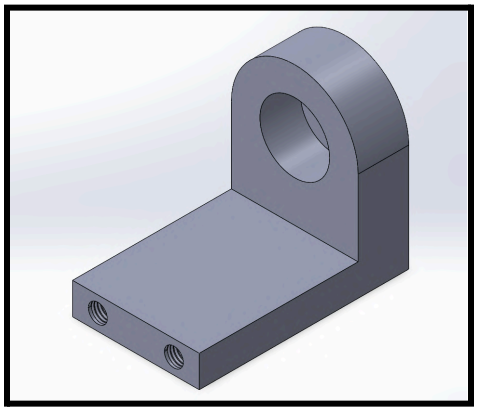
- **CNC fibre laser cutting:** The flat bottom-frame strip is cut from stainless steel sheet, including the long edges and end edges.
- **No bending:** The part remains a flat strip, so there are no bend lines or forming operations.
- **Edge finishing:** The long edges and end edges are deburred and lightly rounded so the strip fits cleanly against the side frames and front frame.
- **Welding fixture positioning:** The bottom frame is placed in a welding fixture with the side frames and front frame. The fixture keeps the heating-frame structure square before welding.
- **TIG tack welding:** Tack welds are placed at the corner/contact points first to hold the bottom frame in position.
- **TIG seam welding:** Weld seams are applied along the contact edges between the bottom frame, side frames, and front frame to create the final welded frame structure.
- **Weld finishing:** The welded edges are cleaned, deburred, brushed, and passivated after welding.

Important manufacturing features:
The bottom frame is cut as a flat strip and becomes the lower support member of the welded heating-frame assembly. The weld seams are placed along the contact edges

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with the side and front frame parts. The welding fixture is important because it keeps the frame straight and square during tack welding and final seam welding.

10.8.5 Mounting Brackets for Bottom Heating Block



Material: EN 1.4301 / AISI 304 stainless steel

Justification:
EN 1.4301 is used because the bracket is close to the bottom heating block and must resist heat, oil vapour, humidity, and cleaning. Stainless steel also reduces heat transfer to the surrounding frame compared with aluminium and gives strong threaded fixing points.

Manufacturing Technology:

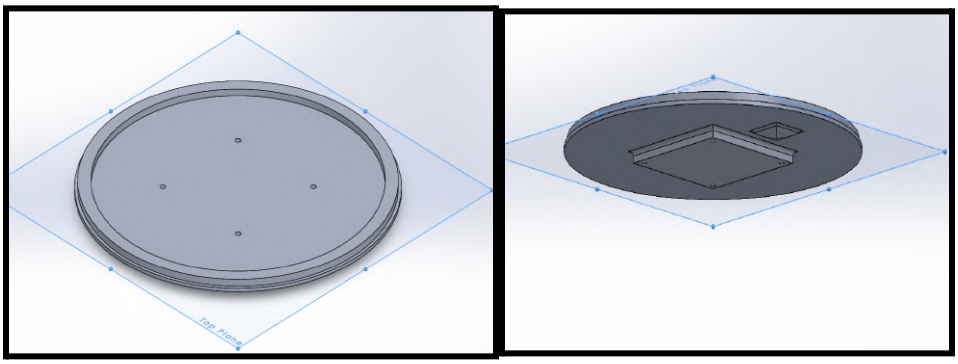
- **Block cutting:** A stainless steel billet or flat bar blank is cut oversized for the base, upright lug, and mounting faces.
- **Setup 1 — base machining:** The blank is clamped from the side faces. The bottom reference face, rectangular base profile, top base surface, and front face are CNC milled.
- **Threaded front holes:** The two front holes are drilled and tapped from the front face. These threads are used to fasten the bracket to the heating-block support frame or adjustment plate.
- **Setup 2 — upright lug machining:** The part is rotated and clamped on the finished base face. The upright lug profile, rounded top shape, and side faces are milled.
- **Large support hole:** The large hole through the upright lug is drilled and bored from the side face. This hole is used for the support pin, spacer, or fixing shaft that holds the heating block position.
- **Finishing:** The large lug hole, front threaded holes, base edges, lug edges, and outer profile are deburred. The part is brushed and passivated after machining.

Important manufacturing features:
The bottom face is the main datum because it controls how the bracket sits under the conveyor. The large lug hole is the critical alignment feature because it positions the heating block support point. The front threaded holes are machined perpendicular to the front face so the fixing screws enter straight and clamp the bracket securely.

10.9. Rotating Plate

10.9.1. Aluminum Plate

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Material: Clear anodised EN AW-6061 T6 aluminium plate

Justification:
EN AW-6061 T6 is used because the rotating plate is supported by a 5 kg load cell, so the plate must stay lightweight. Aluminium also gives enough stiffness for the rotating support surface. Clear anodising improves corrosion resistance, cleanability, and surface protection where cooked eggs may contact the plate.

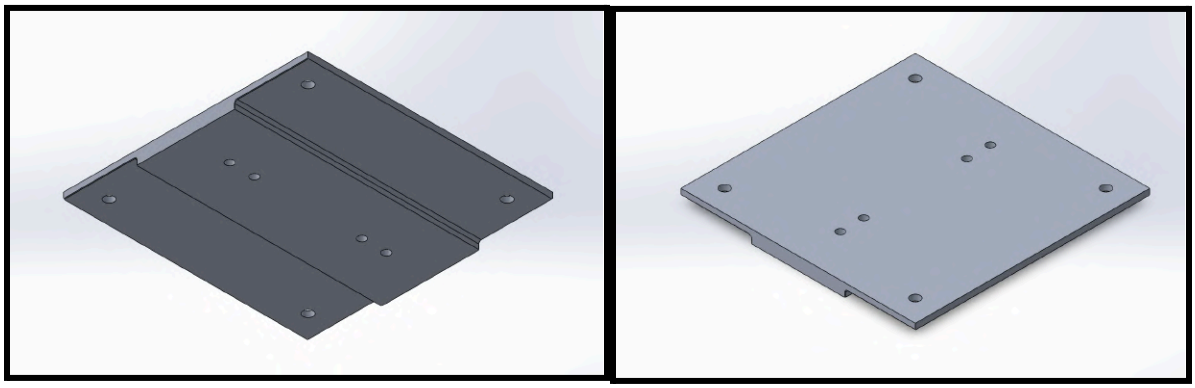
Manufacturing Technology:

- **CNC fibre laser cutting / saw cutting::** A circular aluminium blank is cut slightly oversized from plate material. The blank includes extra allowance for final turning of the outside diameter.
- **Setup 1 — top-side turning:** The circular blank is clamped on the outer diameter and underside. The top serving surface, outer diameter, raised rim, and circular groove/step are turned concentrically.
- **Setup 2 — bottom-side turning:** The plate is flipped and clamped on the finished outer diameter using soft jaws. The bottom reference face is turned flat so the plate sits level on the bearing and load-cell support.
- **Underside CNC milling:** The finished top surface is protected using a vacuum fixture or soft support. The square mounting pocket, underside support pad, rectangular sensor cut-out, and curved microswitch actuator tab are milled on the underside.
- **Microswitch tab finishing:** The curved actuator tab is rounded on its contact edge so it presses the microswitch lever smoothly during rotation.
- **Mounting holes and countersinks:** The mounting holes are drilled through the plate. The countersinks are machined from the top side so flat-head screws sit flush with the egg-contact surface.
- **Finishing:** The outer rim, circular groove, mounting holes, countersinks, underside pocket, rectangular cut-out, and microswitch tab are deburred. The part is then clear anodised.

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Important manufacturing features:
The top serving surface is protected from clamp marks because it may contact cooked eggs. The outer rim and circular groove are turned from the same axis so the plate rotates concentrically. The underside pocket and microswitch tab are machined after turning so they align with the rotating plate position. The countersinks are placed on the top face to keep screw heads flush with the egg-contact surface.

10.9.2. Top Load Cell Plate



Material: Clear anodised EN AW-6061 T6 aluminium plate

Justification:
EN AW-6061 T6 is used because the plate is part of the weighing structure and must stay lightweight so the 5 kg load cell is not overloaded. It also provides enough stiffness for supporting the rotating plate and bearing interface.

Manufacturing Technology:

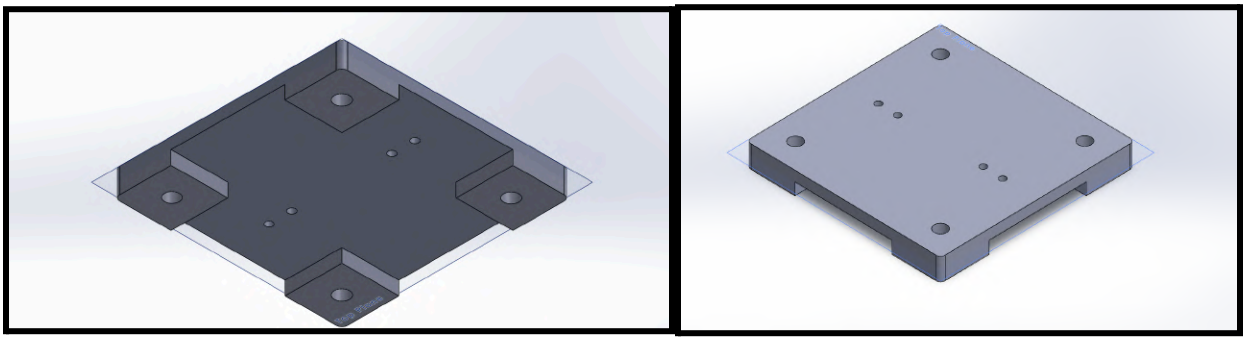
- **CNC fibre laser cutting / saw cutting::** A rectangular aluminium blank is cut slightly oversized, including allowance for the final outer profile and front notch.
- **Setup 1 — top-side machining:** The blank is clamped from the outer side stock. The top reference face, final outer profile, straight edges, front notch, and mounting-hole positions are CNC machined from the top side.
- **Mounting holes and counterbores:** The screw holes are drilled from the top side. Required counterbores are machined on the same side so screw heads or washers sit flat against the mounting surface.
- **Setup 2 — underside pocketing:** The part is flipped and held on the finished top face using a vacuum fixture or soft support. The underside pocket is milled while leaving the central raised rib and thicker mounting pads as part of the plate.
- **Bottom reference face:** The lower contact areas are face milled so the plate sits flat on the load cell interface.

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- **Finishing:** The mounting holes, counterbores, front notch, underside pocket, raised rib, and outer edges are deburred. The plate is then clear anodised.

Important manufacturing features:
The top face is the main datum because it supports the rotating plate and bearing parts. The underside pocket removes weight, while the central rib and mounting pads remain to keep stiffness around the load path. The counterbores are placed only where screw heads or washers need a flat seating surface.

10.9.3. Bottom Load Cell Plate



Material: Clear anodised EN AW-6082 T6 aluminium plate

Justification:
EN AW-6082 T6 is used because the base plate must support the load cell, bearing structure, rotating plate assembly, and motor-side forces with high stiffness. Its mass is below the load cell, so it does not affect the 5 kg measured load, but it must reduce vibration and movement during rotation.

Manufacturing Technology:

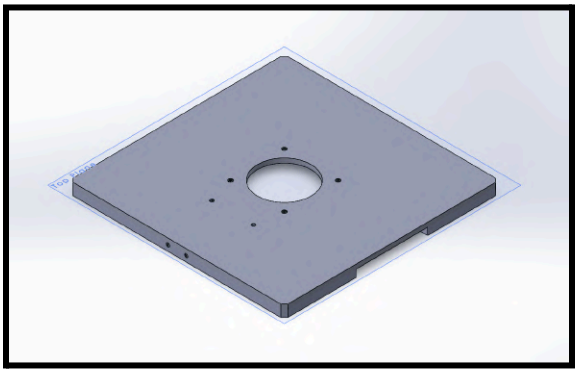
- **CNC fibre laser cutting / saw cutting:** An aluminium plate blank is cut slightly oversized. This step only prepares the raw blank. The final profile, holes, counterbores, threads, reference faces, underside pocket, ribs, and raised pads are produced by CNC machining afterwards.
- **Setup 1 — top-side machining:** The blank is clamped on the outer edges. The top reference surface, final square outer profile, side notches, corner radii, and mounting-hole positions are CNC machined from the top side.
- **Mounting holes and counterbores:** The holes for the load cell, bearing support, motor-side parts, and frame fixing are drilled from the top side. Required counterbores are machined on the same side so screw heads and washers sit flat.

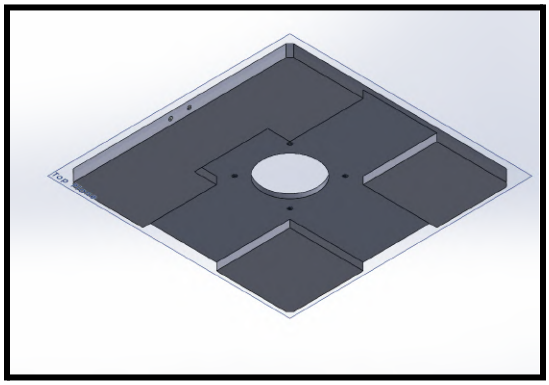
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- **Setup 2 — underside pocketing:** The part is flipped and held on the finished top face using a vacuum fixture or soft support. The underside pocket is milled while leaving the perimeter rim, corner support feet, and central raised support area.
- **Bottom reference areas:** The lower contact pads are face milled so the base plate sits flat on the machine frame and does not twist the load cell structure.
- **Finishing:** The mounting holes, counterbores, side notches, underside pocket, raised pads, and outer edges are deburred. The part is then clear anodised.

Important manufacturing features:
The top surface is the main datum because it controls the load cell and rotating assembly position. The underside pocket reduces weight, while the perimeter rim, corner feet, and central support area remain to keep the plate stiff. The counterbores are placed only where screw heads or washers need a flat seating surface.

10.9.4. Base Plate





Material: Clear anodised EN AW-6082 T6 aluminium plate

Justification:
EN AW-6082 T6 is used because the common base plate supports the rotating plate subsystem, load cell structure, motor housing, and frame connections while keeping the machine weight reasonable. It also provides enough stiffness to reduce vibration during rotation.

Manufacturing Technology:

- **CNC fibre laser cutting / saw cutting:** An aluminium plate blank is cut slightly oversized. This step only prepares the raw blank. The final profile, holes,

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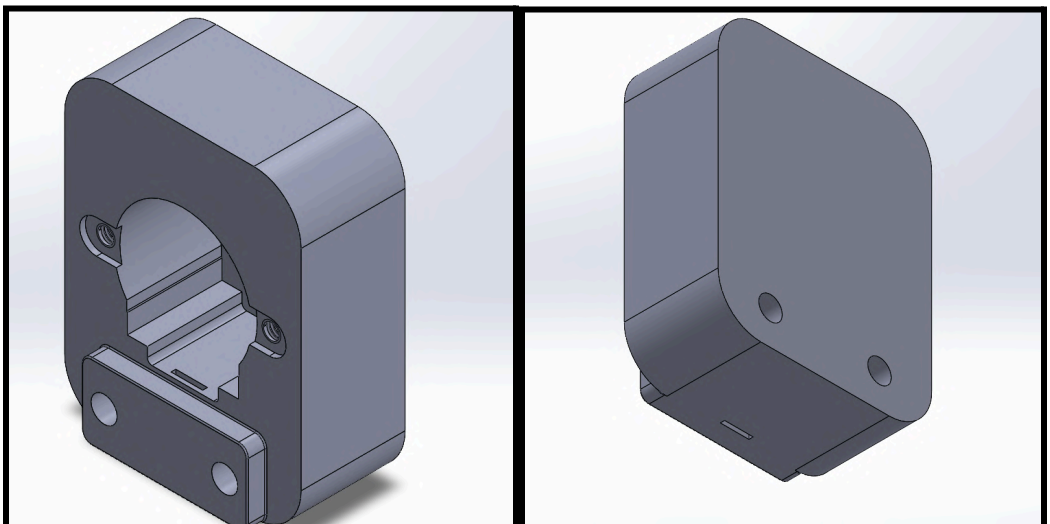
counterbores, threads, reference faces, underside pocket, ribs, and raised pads are produced by CNC machining afterwards.

- **Setup 1 — top-side machining:** The blank is clamped on the outer edges. The top reference face, final square profile, rounded corners, side notches, central through-hole, and top mounting-hole positions are CNC machined from the top side.
- **Top threaded holes and counterbores:** The top holes for the load cell support, motor housing, and bearing-related parts are drilled and thread-formed from above. Required counterbores are machined on the top face so screw heads or washers sit flat.
- **Setup 2 — side threaded holes:** The plate is indexed or re-clamped on its finished top/bottom faces. The side holes are drilled and thread-formed perpendicular to the side faces so the common base plate can be screwed into the surrounding frame.
- **Setup 3 — underside pocketing:** The part is flipped and held on the finished top face using a vacuum fixture or soft support. The underside pocket is milled while leaving the perimeter rim, central support area, raised pads, and ribs for stiffness.
- **Bottom reference areas:** The lower contact pads are face milled so the plate sits flat inside the main frame.
- **Finishing:** The central hole, top threaded holes, side threaded holes, counterbores, underside pocket, raised pads, ribs, side notches, and outer edges are deburred. The part is then clear anodised.

Important manufacturing features:
The top face is the main datum because it controls the load cell, motor housing, and rotating assembly position. The central through-hole is machined from the same top-side setup so it stays aligned with the mounted rotating parts. The side threaded holes are machined perpendicular to the side faces because they connect the plate to the main frame. The underside pocket removes weight, while the perimeter rim, ribs, and raised pads keep the plate stiff.

10.9.5. Stepper Motor Housing

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Material: Clear anodised EN AC-43500 / EN AC-AlSi10MnMg cast aluminium alloy

Justification:
EN AC-43500 is used because the housing must hold the stepper motor and gearbox accurately while keeping the rotating plate subsystem lightweight. Aluminium also helps dissipate heat from the motor area.

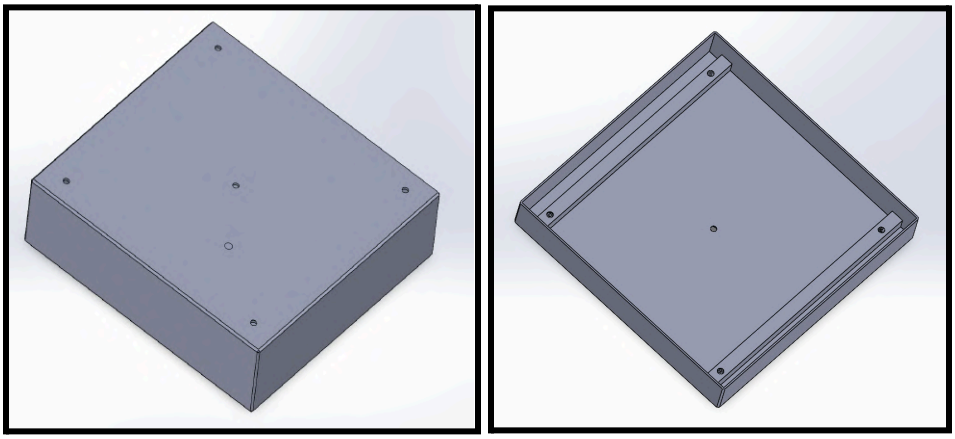
Manufacturing Technology:

- **High-pressure die casting:** The main housing body is cast with the rounded outer casing, front motor face, rear face, side walls, bottom wall, lower mounting flange, and internal motor cavity.
- **Parting line and draft:** The casting parting line is placed around the housing mid-plane. Draft is applied to the outer walls, cavity walls, screw bosses, and flange faces so the housing can be released from the die.
- **Core inserts and side slides:** A fixed core insert forms the internal motor/gearbox cavity, and another core forms the rough front circular opening. Side slides form the side recesses and side-access features that cannot release in the main die-opening direction.
- **CNC reference machining:** The front motor mounting face, rear reference face, bottom mounting face, and lower flange faces are milled after casting so the motor housing sits accurately on the common base plate.
- **Front opening and internal seats:** The front circular motor/gearbox opening is bored to final size from the front face. Internal seating surfaces are pocket milled where the motor and gearbox sit inside the housing.
- **Holes and threads:** The front motor mounting holes are drilled and thread-formed from the front face. The lower flange holes are drilled as clearance holes and counterbored from the mounting side so the screw heads sit flat on the flange.
- **Bottom cable slot:** The small rectangular bottom slot is milled after casting to provide cable clearance from the motor area.
- **Finishing:** The circular opening, threaded holes, counterbores, bottom slot, internal cavity edges, flange edges, and outer housing edges are deburred. The part is clear anodised after machining.

Important manufacturing features:
The front motor face is the main datum because it controls motor and gearbox alignment. The bottom mounting face and lower flange control how the housing sits on the common base plate. The front threaded holes are machined after casting so the motor screws align accurately, while the flange counterbores are placed on the mounting side for stable screw seating.

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10.9.6. Bottom Base for Rotating Plate



Material: EN 1.4301 / AISI 304 stainless steel sheet and flat bar

Justification:
EN 1.4301 is used because the base is inside the food machine and must resist oil, heat, humidity, and cleaning. A welded hollow sheet-metal box is used instead of a solid block to reduce weight and material cost, while internal flat bars provide enough thread depth for the M8 mounting holes.

Manufacturing Technology:

- **CNC laser cutting:** The top plate and four side wall panels are cut from stainless steel sheet. The outer profiles are cut flat before welding.
- **Flat bar cutting:** Two stainless steel flat bars are cut to length. These bars are placed under the top plate where the M8 threaded holes are needed.
- **Outer box welding:** The four side panels are positioned perpendicular to the top plate in a welding fixture. TIG weld seams are applied along the top-to-side panel contact edges and vertical corner edges to create the hollow box frame.
- **Internal support bar welding:** The box is flipped, and the two flat bars are clamped against the underside of the top plate. They are welded directly below the future M8 hole positions to create thick threaded zones.
- **Hole machining after welding:** The welded frame is clamped right-side up. The four corner M8 holes and central through-holes are spot drilled after welding so their positions remain accurate even if welding causes slight distortion.
- **Threaded holes:** The M8 core holes are drilled vertically through the top sheet and into the welded internal flat bars, then tapped M8×1.25 to give full thread engagement.
- **Central holes:** The centre holes are drilled through the top sheet only, where clearance or alignment is needed for the mounted components.

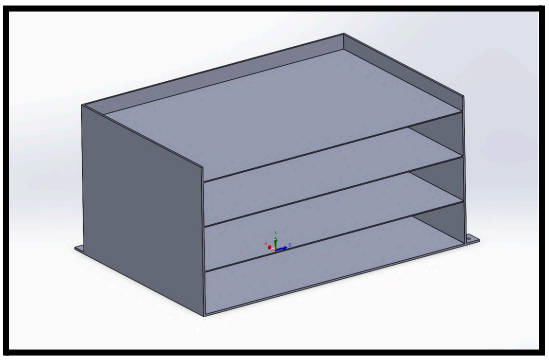
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- **Weld finishing:** The outside corner welds and top-edge weld seams are ground flush. The visible surfaces are brushed for a clean hygienic finish.
- **Final finishing:** The M8 threaded holes, central holes, top edges, side edges, and welded corners are deburred and passivated.

Important manufacturing features:
The internal flat bars are positioned directly under the M8 holes because the thin sheet alone would not provide enough thread depth. The mounting holes are drilled and tapped after welding so the hole positions match the final welded frame, not the pre-weld sheet parts. The weld seams are placed along the outer box edges and are ground smooth so residue cannot collect easily.

10.10. Electrical Box

10.10.1 Electrical Box Housing



Material: EN 1.4301 / AISI 304 stainless steel sheet, with thicker EN 1.4301 mounting tabs

Justification:
EN 1.4301 is used because the electrical housing must be rigid, corrosion-resistant, and easy to clean inside the machine. Sheet metal keeps the housing lighter than a solid block, while thicker mounting tabs provide stronger fixing points to the machine frame.

Manufacturing Technology:

- **CNC fibre laser cutting:** The back panel, side walls, top and bottom panels, internal shelves, and external mounting tabs are cut from flat stainless steel sheet. Any cable openings, ventilation holes, and fastening holes are also cut while the sheets are flat.
- **Mounting tab holes:** The external tabs are drilled and countersunk before welding. The countersinks are made on the outer face so flat-head screws sit flush when the housing is fixed to the machine frame.
- **Main shell welding:** The side walls, top panel, bottom panel, and back panel are aligned in a welding fixture. TIG weld seams are placed along the rear vertical edges and the top/bottom contact edges to form the open-front housing box.
- **Internal shelf welding:** The horizontal shelves are positioned inside the housing and TIG welded along their contact lines with the side walls and back panel. These weld lines hold the shelves and also stiffen the housing.

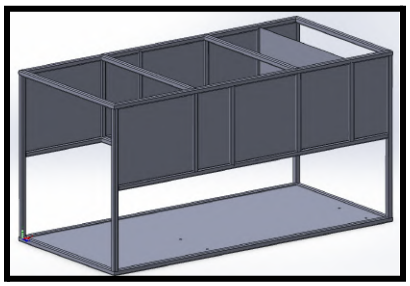
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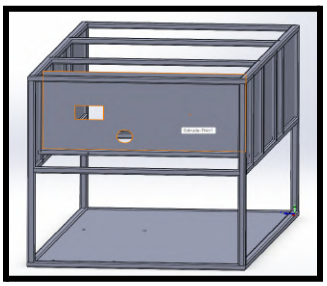
- **Mounting tab welding:** The prepared tabs are welded to the outside lower side edges of the housing. The tab welds are placed along the tab-to-wall contact edges so the screw loads transfer into the side walls.
- **Weld finishing:** External welds are ground smooth where they are visible or where the housing must sit against the frame. Internal shelf welds are cleaned but not over-ground, so shelf strength is not reduced.
- **Edge finishing:** All sheet edges, shelf fronts, cable openings, ventilation holes, countersinks, and welded corners are deburred so wires cannot be damaged during installation.
- **Surface finishing:** The visible outer surfaces are brushed, and the welded areas are passivated to restore corrosion resistance.

Important manufacturing features:
The internal shelves are welded after the main box is squared in a fixture so the housing does not twist. The mounting tabs are made from thicker stock because they carry the fixing loads. All cable openings and shelf front edges must be rounded because sharp stainless edges could cut insulation during assembly or maintenance.

10.11 Main Frame

10.11.1 Main Frame housing





Material: EN 1.4301 / AISI 304 stainless steel square hollow sections and stainless steel sheet

Justification:
EN 1.4301 is used because the frame is inside a food-machine environment and must resist heat, oil vapour, humidity, and cleaning. Square hollow sections provide the main stiffness with low weight, while sheet-metal panels close the frame and support internal components.

Manufacturing Technology:

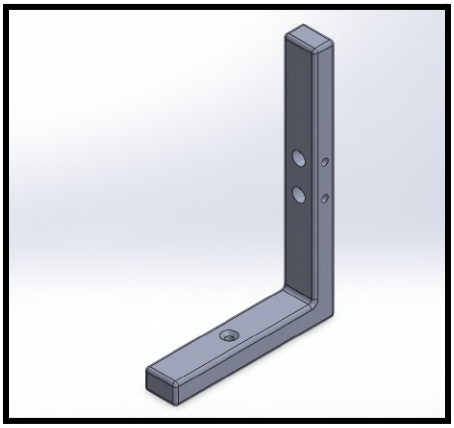
- **Tube cutting:** The stainless steel square hollow sections are cut to length for the vertical posts, bottom rails, and top rails. The tube ends are cut square or mitred depending on the frame corner connection.
- **Sheet laser cutting:** The base plate, side panels, and internal divider panels are laser cut from stainless steel sheet. The square cut-out, circular hole, cable openings, and mounting holes are cut while the panels are still flat.

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- **Base plate countersinks:** The base-plate mounting holes are countersunk from the top side so flat-head screws or levelling feet sit flush and do not interfere with internal parts.
- **Frame hole machining:** Small fastening holes are drilled or tube-laser cut into the square tubes where external covers, cable clips, or internal brackets are later attached.
- **Frame welding:** The cut square tubes are held in a welding fixture and TIG welded at the corner joints and tube intersections. The fixture keeps the frame square during welding.
- **Sheet panel welding:** The base plate, side panels, and internal divider panels are positioned against the tube frame. TIG tack welds are added first, then short stitch welds are used along the panel-to-tube contact edges to reduce sheet distortion.
- **Weld finishing:** External tube-frame welds are ground smooth where the frame is visible or where panels must sit flush. Internal welds are cleaned but not over-ground so the frame strength is maintained.
- **Edge finishing:** The laser-cut panel edges, square cut-out, circular hole, cable openings, countersinks, and drilled tube holes are deburred to remove sharp edges.
- **Surface finishing:** The visible stainless surfaces are brushed, and the welded areas are passivated after welding and grinding.

Important manufacturing features:
The square hollow tubes form the load-bearing skeleton, so their weld joints are made in a fixture to keep the frame straight. The sheet panels are stitch welded instead of fully seam welded to reduce heat distortion. The square and circular cut-outs are made before welding because they are easier to cut accurately while the sheet is flat. All cable and pipe openings are deburred so wiring, tubing, and cleaning access are not damaged by sharp edges.

10.11.2. L bracket for servo motor



Material: EN 1.4301 / AISI 304 stainless steel sheet, 2.0 mm

Justification:
EN 1.4301 is used because the bracket is inside the machine and must resist humidity, oil vapour, heat, and cleaning. Stainless steel also provides enough stiffness for a small structural mounting bracket.

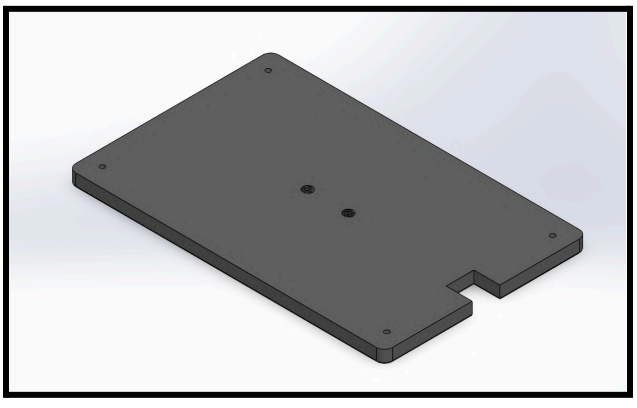
Manufacturing Technology:

- **CNC fibre laser cutting:** The flat sheet blank is cut first, including the outer profile, base mounting hole, vertical mounting holes, side holes, and bend reliefs.
- **Hole cutting:** All circular holes are cut while the sheet is flat so their positions stay accurate before bending.
- **Press brake bending:** The sheet is bent 90° along the bend line between the horizontal base flange and the vertical upright flange. The inside radius is created by the press brake tool.
- **Bend direction:** The vertical flange is bent upward from the base flange. The base hole remains on the horizontal mounting side, while the vertical holes are used for fastening the supported part.
- **Edge finishing:** The outer edges, hole edges, bend reliefs, and flange corners are deburred and lightly rounded.
- **Surface finishing:** The visible surfaces are brushed, and the part is passivated after cutting and bending.

Important manufacturing features:
The bend line controls the 90° angle between the base flange and upright flange. The holes are cut before bending because they are easier to position accurately in the flat blank. The vertical holes must stay aligned after bending because they locate the mounted component.

10.12 Display Casing

10.12.1 Plastic Back Plate



Display Back Plate
Material: ABS-HI plastic with brass threaded inserts

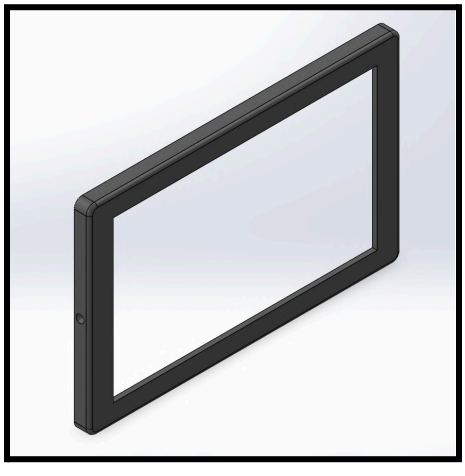
Justification:
ABS-HI is used because the back plate is a non-food-contact display support part that must stay lightweight, impact-resistant, and dimensionally stable. Brass threaded inserts are used for the display mounting points because repeated screw tightening would damage plastic threads.

Manufacturing Technology:

- **Injection moulding:** The back plate is moulded as one plastic part, including the flat rectangular plate, rounded corners, side relief notch, rear stiffening ribs, and local screw bosses.
- **Mould parting line:** The parting line is placed around the outer edge of the plate so the main flat mounting face stays clean.
- **Core pins for corner holes:** The four corner fixing holes are formed with core pins through the plate thickness. These holes are used to fasten the back plate to the display housing or frame.
- **Heat-set threaded inserts:** Brass inserts are pressed into the two central display mounting bosses after moulding. These create stronger threads for attaching the display module or bracket.
- **Relief notch:** The side notch is formed directly in the mould cavity to give clearance for nearby cables, connectors, or housing features.
- **Gate position:** The injection gate is placed on the rear side of the plate so no gate mark appears on the visible or functional mounting face.
- **Finishing:** The outer edge, corner holes, notch edges, insert openings, and parting-line flash are trimmed and deburred.

Important manufacturing features:
The central bosses are the critical features because they hold the threaded inserts for the display screws. The corner holes are moulded through the plate so the back plate can be fixed to the cover or frame. The main mounting face must remain flat so the display sits evenly without bending or twisting.

10.12.2 Front Frame for Casing



Material: ABS-HI plastic

Justification:
ABS-HI is used because the front cover is a visible user-facing part that needs good impact resistance, a clean surface finish, and enough stiffness around the display opening. It is suitable because the part is not

exposed to direct food contact or high heating surfaces.

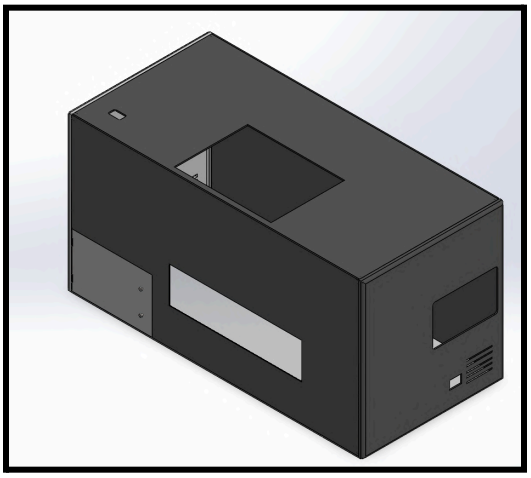
Manufacturing Technology:

- **Injection moulding:** The front cover is moulded as one plastic bezel, including the rectangular frame, rounded outer corners, inner display opening, side wall, and side screw hole boss.
- **Mould parting line:** The parting line is placed around the outer side edge of the bezel so the visible front face remains clean.
- **Core insert for display opening:** The large rectangular display window is formed directly by a core insert in the mould. This keeps the opening shape accurate and avoids cutting the window after moulding.
- **Core pin for side hole:** The side mounting hole is formed by a side core pin or slide because the hole direction is perpendicular to the main mould opening direction.
- **Rear-side bosses/ribs:** Local thickened areas are moulded on the rear side around screw positions and frame edges to support the display and prevent the bezel from flexing.
- **Gate position:** The injection gate is placed on the rear side of the cover so no gate mark appears on the visible front surface.
- **Finishing:** The inner display opening, outer edges, side hole, and parting-line flash are trimmed and deburred after moulding.

Important manufacturing features:
The front face is the visible user-facing surface, so gate marks, ejector marks, and parting lines are kept away from it. The display opening is the critical feature because it must fit the screen cleanly. The side hole is made with a side slide/core pin because it cannot be formed in the normal mould opening direction.

10.13 Main Case/Chassy

10.13.1 All Chassy Parts are made in the Same Manner



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Material: ABS-HI plastic sheet panels, with purchased glass window insert

Justification:
ABS-HI is used because the outer case must be lightweight, impact-resistant, and easy to clean. It is suitable for the external housing because it is not a direct food-contact or heating surface.

The glass window is bought separately because it needs optical clarity and scratch resistance.

Manufacturing Technology:

- **CNC routing of plastic panels:** The six case sides are cut from ABS-HI sheet: top panel, bottom panel, front panel, back panel, left side panel, and right side panel.
- **Panel cut-outs:** The front viewing opening, top square opening, side display opening, ventilation slots, button/sensor holes, and service openings are CNC routed while the panels are flat.
- **Corner radii and edge rounding:** The outside panel corners and visible edges are rounded with a router radius tool or edge-finishing tool so the case has safer, smoother edges.
- **Panel joining:** The six plastic panels are aligned in an assembly fixture and joined along the inside corner seams using plastic solvent bonding or plastic welding. The outside faces are kept clean and free from visible bonding marks.
- **Glass window installation:** The glass is a purchased part and is installed into the front rectangular opening using a rubber gasket or heat-resistant adhesive frame. The glass is not manufactured in-house.
- **Ventilation and access features:** The ventilation slots are placed on the side panel to allow airflow for the electronics area. The display and button cut-outs are made before assembly so their positions stay accurate.
- **Finishing:** All panel edges, cut-out edges, ventilation slots, glass opening, and display opening are deburred and lightly rounded before final assembly.

Important manufacturing features:
Each cut-out is made while the panel is flat because it is more accurate than cutting after the box is assembled. The glass opening must have a clean, flat edge so the gasket or adhesive frame seals properly. The corner seams are joined from the inside so the outside case surface stays clean and presentable.

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